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Software Service Documentation

Amazon Web Services Performance and Sizing

SampleManager 12.2 SP1 – Web Client

Document Control

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1 Document Overview

This document describes performance testing carried out against SampleManager 12.2SP1 web architecture and the resultant recommendations. SampleManager 12.2SP1 web client performance data was generated and used as a baseline. Tests runs were completed against Oracle database only.

The information is presented in the following sections:

1) Recommendations
2) System Performance Monitoring
3) User Experience Timings
4) Network Performance

2 Recommendations

SampleManager web system is a four-tier application, consisting of a client browser machine, web server, application server and database server. This section summarizes the other sections of this document to pull out some high level recommendations for the hardware and networking requirements for a SampleManager web system. As the testing was carried out in Amazon Web Services (AWS) environment, virtual machines used for hosting different tiers of SampleManager web system were choosen from pre-configured AWS EC2 instance types.

2.1 Client

The client requires the following hardware:

	Minimum Specification
CPU	2.5 GHz processor
RAM	4 GB
Display	1400 × 1050 resolution with support for Microsoft® DirectX®

2.2 Web Server

At a minimum, the SampleManager web server on AWS requires the following hardware:

- Base System 2 core processors
  - Allow 2 cores for every 10 concurrent connections when the total number of core processors is 8 or less. When the total number of cores is 16 or greater, only 1 core is recommended for every extra 10 concurrent connections.
- Base RAM
  - Size of RAM should not be lower than RAM recommendation for SampleManager application server. (2GB + 125 MB for each concurrent connection)
- 1 GB hard disk space (760 MB for install files)

Example recommendations based on the load testing performed are:

AWS EC2 Instance Type	m5-large	m5-xlarge	m5-2xlarge	m5-4xlarge
CPU	2 virtual core processors	4 virtual core processors	8 virtual core processors	16 virtual core processors
Number of Concurrent Connections	0-10	10-20	20-45	45-180

Note: Server requirements are given for a SampleManager web server only.

Assigned virtual core processors are more critical than just RAM of the server which impacts overall performance of the SampleManager web server concurrent connections.

2.3 Networking

Please see the networking section of this document for more information. Latency has a gradual effect on performance.

2.3.1 Client Browser to Web Server

Lower network latency is desired for optimal performance.

2.3.2 Web Server to Application Server

Less than 1ms network latency.

3 System Performance Monitoring

3.1 Introduction

This section describes the testing work carried out to determine the resource requirements for the SampleManager web server system.

3.2 Setup

3.2.1 Introduction

Differernt types of EC2 machine instances in an Amazon Web Services (AWS) environment were used to carry out all testing for the SampleManager web client. JMeter is used as a load testing tool for analyzing and measuring the capability of SampleManager web client. The following section describes the scripts developed along with the machine configurations.

3.2.2 Client Script

JMeter HTTP Header Manager was used to send User-Agent header to mimic browser type (Chrome) on browser client machine.

The following steps were automated with JMeter to simulate user actions with JMeter virtual users. The collected data was used to measure server side performance under different load conditions. 1-180 users load testing was recorded and analyzed. More realistic user load parameter was used (e.g. for each user there will be a certain number of seconds 'thinking' time for each step shown below to ensure that the number of concurrent users are all logged on during the testing).

1	SampleManager 12.2SP1 web client start page
2	Login to SampleManager web client
3	Navigate to Personal Folder
4	Navigate to Analysis Folder
5	Navigate to Data Trail Folder
6	Logout SampleManager web client

3.2.3 Test Cases

SampleManager Web Server	
AWS EC2 Instance Type	m5-large
CPU	2 Virtual Core Processors
RAM	8 GB RAM
Concurrent Connections Supported	10

Label	# Http Requests	Avg (ms)	Median (ms)	90% Line (ms)	Min (ms)	Max (ms)	Error %	Throughput /Sec	Received KB/sec	Sent KB/sec
Start page	5	55	56	56	55	57	0.00%	0.98932	16.28	0.5
Login	5	4545	4574	4577	4297	4708	0.00%	0.52203	434.5	0.76
Personal Folder	5	2732	2645	3029	2286	3084	0.00%	0.63299	526.37	0.5
Analysis	5	2784	3028	3072	1828	3096	0.00%	0.57864	481.17	0.45
Result Trail	5	2447	2668	2762	1922	2818	0.00%	0.52731	438.49	0.41
Logout	5	105	107	113	93	115	0.00%	0.65334	10.56	0.84
TOTAL	30	2111	2617	4572	55	4708	0.00%	0.41637	233.14	0.39

Label	# Http Requests	Avg (ms)	Median (ms)	90% Line (ms)	Min (ms)	Max (ms)	Error %	Throughput /Sec	Received KB/sec	Sent KB/sec
Start Page	10	272	58	143	56	2011	0.00%	0.83886	13.8	0.43
Login	10	14990	14340	17938	11795	18160	0.00%	0.42369	352.66	0.62
Personal Folder	10	8326	8619	9283	3180	10591	0.00%	0.69416	577.23	0.54
Analysis	10	9836	10506	10833	3739	11088	0.00%	0.47771	397.25	0.37
Result Trail	10	8669	9309	9674	3733	9885	0.00%	0.37732	313.76	0.3
Logout	10	513	174	392	101	3261	0.00%	0.43437	7.02	0.56
TOTAL	60	7101	8940	13588	56	18160	0.00%	0.5662	317.04	0.53

Label	# Http Requests	Avg (ms)	Median (ms)	90% Line (ms)	Min (ms)	Max (ms)	Error %	Throughput /Sec	Received KB/sec	Sent KB/sec
Start Page	15	156	58	268	55	683	0.00%	0.98245	16.17	0.5
Login	15	10789	8191	18838	4497	20810	6.67%	0.41644	323.61	0.59
Personal Folder	15	11555	11372	15001	7194	15107	13.33%	0.33023	238.1	0.22
Analysis	15	10574	12358	14514	4341	15001	13.33%	0.38511	277.67	0.26
Result Trail	15	4809	4978	6317	2212	6857	0.00%	0.46819	389.33	0.37
Logout	15	340	112	320	92	2052	0.00%	0.58074	9.39	0.74
TOTAL	90	6371	5690	14728	55	20810	5.56%	0.81222	417.38	0.72

SampleManager Web Server	
AWS EC2 Instance Type	m5-xlarge
CPU	4 Virtual Core Processors
RAM	16 GB RAM
Concurrent Connections Supported	20

Label	# Http Requests	Avg (ms)	Median (ms)	90% Line (ms)	Min (ms)	Max (ms)	Error %	Throughput /Sec	Received KB/sec	Sent KB/sec
Start Page	15	145	57	145	56	637	0.00%	0.99056	16.3	0.5
Login	15	5027	4900	5497	4225	6825	0.00%	0.68465	569.9	1
Personal Folder	15	3363	3026	4000	2457	5695	0.00%	0.67078	557.84	0.53
Analysis	15	3866	3443	5460	1999	6024	0.00%	0.69567	578.53	0.55
Result Trail	15	2467	2398	2912	1749	2970	0.00%	0.75184	625.25	0.59
Logout	15	129	112	133	101	331	0.00%	0.83804	13.54	1.07
TOTAL	90	2500	2575	5173	56	6825	0.00%	1.05111	588.6	0.98

Label	# Http Requests	Avg (ms)	Median (ms)	90% Line (ms)	Min (ms)	Max (ms)	Error %	Throughput /Sec	Received KB/sec	Sent KB/sec
Start Page	20	139	69	249	56	857	0.00%	0.99651	16.4	0.51
Login	20	7923	5052	12898	4322	15122	0.00%	0.60772	505.87	0.89
Personal Folder	20	6409	5960	9382	3050	11161	0.00%	0.61015	507.41	0.48
Analysis	20	4539	4603	5866	2231	8006	0.00%	0.63853	531.01	0.5
Result Trail	20	3123	2856	4379	1904	5081	0.00%	0.77676	645.97	0.61
Logout	20	140	108	180	91	339	0.00%	0.90827	14.68	1.16
TOTAL	120	3712	3260	8539	56	15122	0.00%	1.22232	684.47	1.14

Label	# Http Requests	Average (ms)	Median (ms)	90% Line (ms)	Min (ms)	Max (ms)	Error %	Throughput /Sec	Received KB/sec	Sent KB/sec
Start Page	25	268	107	495	56	1845	0.00%	0.93131	15.32	0.47
Login	25	11737	6156	25784	3889	27820	12.00%	0.49234	360.86	0.67
Personal Folder	25	9324	7146	15001	2620	15002	32.00%	0.48271	273.36	0.26
Analysis	25	7672	6534	12916	4084	15002	4.00%	0.4637	370.24	0.35
Result Trail	25	4053	2936	6480	1943	7714	0.00%	0.59339	493.47	0.47
Logout	25	232	114	276	92	1207	0.00%	0.67558	10.92	0.87
TOTAL	150	5548	4589	15000	56	27820	8.00%	1.28521	634.45	1.12

SampleManager Web Server	
AWS EC2 Instance Type	m5-2xlarge
CPU	8 Virtual Core Processors
RAM	32 GB RAM
Concurrent Connections Supported	45

Label	# Http Requests	Avg (ms)	Median (ms)	90% Line (ms)	Min (ms)	Max (ms)	Error %	Throughput /Sec	Received KB/sec	Sent KB/sec
Start Page	25	94	58	155	55	284	0.00%	0.99234	16.33	0.51
Login	25	4389	4086	5188	3517	7601	0.00%	0.86275	718.07	1.26
Personal Folder	25	2332	2320	2618	1753	3254	0.00%	0.92057	765.48	0.72
Analysis	25	2114	2092	2377	1706	2671	0.00%	0.8934	742.89	0.7
Result Trail	25	1986	1920	2361	1701	2518	0.00%	0.91699	762.51	0.72
Logout	25	118	103	176	91	218	0.00%	0.98869	15.98	1.27
TOTAL	150	1839	1975	3985	55	7601	0.00%	1.67909	940.15	1.57

Label	# Http Requests	Avg (ms)	Median (ms)	90% Line (ms)	Min (ms)	Max (ms)	Error %	Throughput /Sec	Received KB/sec	Sent KB/sec
Start Page	30	82	54	160	52	223	0.00%	0.99817	16.42	0.51
Login	30	5294	4656	7621	4070	8515	0.00%	0.80066	666.52	1.17
Personal Folder	30	2731	2754	3674	1713	4407	0.00%	0.92254	767.26	0.72
Analysis	30	2826	2749	3831	1661	4121	0.00%	0.93032	773.73	0.73
Result Trail	30	2318	1960	3557	1630	3931	0.00%	0.95727	796.14	0.75
Logout	30	133	106	182	94	412	0.00%	1.02337	16.54	1.31
TOTAL	180	2230	2052	4581	52	8515	0.00%	1.83561	1027.98	1.71

Label	# Http Requests	Avg (ms)	Median (ms)	90% Line (ms)	Min (ms)	Max (ms)	Error %	Throughput /Sec	Received KB/sec	Sent KB/sec
Start Page	35	510	55	1500	53	3511	0.00%	0.99381	16.35	0.51
Login	35	9538	9683	16290	4112	20392	0.00%	0.78521	653.66	1.14
Personal Folder	35	5628	5185	9043	1814	11777	0.00%	0.761	632.91	0.6
Analysis	35	4197	3002	6916	2101	11059	0.00%	0.76279	634.4	0.6
Result Trail	35	2816	2303	4971	1653	6484	0.00%	0.88982	740.04	0.7
Logout	35	119	106	139	88	332	0.00%	0.93065	15.04	1.19
TOTAL	210	3801	2577	9735	53	20392	0.00%	1.93285	1082.43	1.81

Label	# Http Requests	Avg (ms)	Median (ms)	90% Line (ms)	Min (ms)	Max (ms)	Error %	Throughput /Sec	Received KB/sec	Sent KB/sec
Start Page	45	133	112	236	52	373	0.00%	0.99798	16.42	0.51
Login	45	5332	4561	7505	3424	12461	0.00%	0.78262	651.45	1.14
Personal Folder	45	3742	3385	6406	1899	7450	0.00%	0.81052	674.04	0.64
Analysis	45	3685	3294	5657	1779	10646	0.00%	0.81336	676.4	0.64
Result Trail	45	3299	2235	5860	1603	8939	0.00%	0.82407	685.31	0.65
Logout	45	181	104	348	87	806	0.00%	0.90792	14.67	1.16
TOTAL	270	2729	2318	5910	52	12461	0.00%	2.30874	1292.84	2.16

Label	# Http Requests	Avg (ms)	Median (ms)	90% Line (ms)	Min (ms)	Max (ms)	Error %	Throughput /Sec	Received KB/sec	Sent KB/sec
Start Page	50	405	102	992	55	4371	0.00%	0.91961	15.13	0.47
Login	50	9674	6321	19229	4058	26620	22.00%	0.66835	434.38	0.78
Personal Folder	50	6814	3325	15001	186	15002	30.00%	0.69232	301.94	0.39
Analysis	50	6533	4644	15001	56	15002	18.00%	0.69484	372.18	0.46
Result Trail	50	4756	2526	15000	56	15003	12.00%	0.70345	411.74	0.44
Logout	50	193	110	360	92	1153	0.00%	0.88998	14.38	1.08
TOTAL	300	4729	2507	15001	55	26620	13.67%	2.17004	810.02	1.72

SampleManager Web Server	
AWS EC2 Instance Type	m5-4xlarge
CPU	16 Virtual Core Processor
RAM	64 GB RAM
Concurrent Connections Supported	180

Label	# Http Requests	Avg (ms)	Median (ms)	90% Line (ms)	Min (ms)	Max (ms)	Error %	Throughput /Sec	Received KB/sec	Sent KB/sec
Start Page	50	92	63	153	53	238	0.00%	0.99894	16.44	0.51
Login	50	4056	3907	4345	3585	7589	0.00%	0.92023	765.92	1.34
Personal Folder	50	1950	1908	2167	1612	2397	0.00%	0.9569	795.7	0.75
Analysis	50	1923	1904	2146	1632	2371	0.00%	0.95418	793.44	0.75
Result Trail	50	1866	1841	2076	1638	2223	0.00%	0.95438	793.61	0.75
Logout	50	121	103	165	88	358	0.00%	0.98645	15.94	1.27
TOTAL	300	1668	1830	3840	53	7589	0.00%	2.56329	1435.26	2.39

Label	# Http Requests	Avg (ms)	Median (ms)	90% Line (ms)	Min (ms)	Max (ms)	Error %	Throughput /Sec	Received KB/sec	Sent KB/sec
Start Page	70	86	57	163	53	211	0.00%	0.99833	16.43	0.51
Login	70	3906	3824	4325	3535	4778	0.00%	0.94243	784.47	1.37
Personal Folder	70	1973	1921	2235	1684	2367	0.00%	0.96238	800.33	0.76
Analysis	70	1943	1888	2296	1617	2539	0.00%	0.96464	802.21	0.76
Result Trail	70	1895	1831	2212	1628	2524	0.00%	0.96246	800.4	0.76
Logout	70	129	107	203	88	387	0.00%	0.98701	15.95	1.27
TOTAL	420	1655	1815	3748	53	4778	0.00%	2.6214	1467.92	2.45

Label	# Http Requests	Avg (ms)	Median (ms)	90% Line (ms)	Min (ms)	Max (ms)	Error %	Throughput /Sec	Received KB/sec	Sent KB/sec
Start Page	90	91	56	168	53	594	0.00%	0.99808	16.42	0.51
Login	90	3991	3964	4352	3493	4817	0.00%	0.9527	793.02	1.39
Personal Folder	90	2001	1960	2258	1663	2573	0.00%	0.9662	803.51	0.76
Analysis	90	1987	1957	2287	1647	2843	0.00%	0.96755	804.63	0.76
Result Trail	90	1902	1854	2216	1607	2754	0.00%	0.96843	805.36	0.76
Logout	90	130	104	191	87	556	0.00%	0.99174	16.03	1.27
TOTAL	540	1684	1839	3881	53	4817	0.00%	2.65592	1487.25	2.48

Label	# Http Reuests	Avg (ms)	Median (ms)	90% Line (ms)	Min (ms)	Max (ms)	Error %	Throughpu t /Sec	Received KB/sec	Sent KB/sec
Start Page	120	92	57	179	53	296	0.00%	0.99791	16.42	0.51
Login	120	3992	3944	4412	3498	4848	0.00%	0.96421	802.62	1.4
Personal Folder	120	2001	1993	2271	1663	2680	0.00%	0.97726	812.72	0.77
Analysis	120	1954	1946	2221	1634	2532	0.00%	0.9776	813.01	0.77
Result Trail	120	1904	1863	2273	1632	2576	0.00%	0.97901	814.18	0.77
Logout	120	138	108	232	88	448	0.00%	0.99686	16.11	1.28
TOTAL	720	1680	1839	3885	53	4848	0.00%	2.71763	1521.84	2.54

Label	# Http Requests	Avg (ms)	Median (ms)	90% Line (ms)	Min (ms)	Max (ms)	Error %	Throughput /Sec	Received KB/sec	Sent KB/sec
Start Page	160	103	60	206	53	437	0.00%	0.99841	16.43	0.51
Login	160	4053	3990	4604	3538	5031	0.00%	0.96839	806.14	1.41
Personal Folder	160	2054	1977	2455	1663	2910	0.00%	0.97892	814.14	0.77
Analysis	160	2005	1955	2402	1628	2988	0.00%	0.98112	815.97	0.77
Result Trail	160	1946	1883	2301	1618	3115	0.00%	0.98229	816.95	0.77
Logout	160	131	104	220	77	450	0.00%	0.99462	16.08	1.28
TOTAL	960	1715	1859	3877	53	5031	0.00%	2.73631	1532.37	2.56

Label	# Http Requests	Avg (ms)	Median (ms)	90% Line (ms)	Min (ms)	Max (ms)	Error %	Throughput /Sec	Received KB/sec	Sent KB/sec
Start Page	180	122	62	271	53	514	0.00%	0.99771	16.42	0.51
Login	180	4395	4202	5436	3616	6393	0.00%	0.96754	805.43	1.41
Personal Folder	180	2372	2237	3139	1649	3641	0.00%	0.97507	810.94	0.77
Analysis	180	2293	2150	3103	1651	3546	0.00%	0.97494	810.83	0.77
Result Trail	180	2159	1982	3055	1633	3762	0.00%	0.97549	811.29	0.77
Logout	180	141	109	221	89	516	0.00%	0.98999	16	1.27
TOTAL	1080	1914	1996	4062	53	6393	0.00%	2.81534	1576.63	2.63

Label	# Http Requests	Avg (ms)	Median (ms)	90% Line (ms)	Min (ms)	Max (ms)	Error %	Throughput /Sec	Received KB/sec	Sent KB/sec
Start Page	190	185	59	195	53	4914	0.00%	0.97479	16.04	0.5
Login	190	5121	4053	4913	3595	25504	7.37%	0.9054	698.35	1.23
Personal Folder	190	5688	2095	15001	55	15002	26.84%	0.92336	518.82	0.54
Analysis	190	5745	2114	15001	53	15003	26.32%	0.93198	527.74	0.54
Result Trail	190	5474	1928	15001	52	15002	25.79%	0.94099	536.92	0.53
Logout	190	531	107	332	88	9285	4.21%	0.9507	14.84	1.17
TOTAL	1140	3791	1924	15000	52	25504	15.09%	2.66248	1110.31	2.14

Based on the information collected above, it was observed that increasing the number of virtual core processors on the web server machine improves total number of concurrent connections for SampleManager web client. The testing Error % > 0.00 happens when the uage of CPU on web server reached >95%, indicating SampleManager web system greatly depends on the power of CPU of web server and the CPU usage is critical to the performance of SampleManager web system. It was also observed that the change of RAM utilization during each testing varied, but not very dramatic as memory utilization of IIS worker process was managed by IIS.

3.2.4 Configuration

SampleManager web system was setup in the typical four tier mode with browser client, web

server, application server and database separated in Amazon Web Services (AWS). Launching SampleManager web client from browser client machine. The machines in this scenario were pre-configured EC2 type virtual machine instances hosted via AWS. The following table summarizes the application server, web server and browser client machines.

	Application Server	Web Server	Browser Client
AWS EC2 Instance Type	m5-4xlarge	m5-nlarge (n=1, x, 2x, 4x)	c5-xlarge
OS	Windows Server 2016 Datacenter 64-bit	Windows Server 2016 Datacenter 64-bit	Windows Server 2019 Datacenter 64-bit
CPU	16 Virtual Processors	2-16 Virtual Processors	4 Virtual Processors
RAM	64 GB	8 - 64GB	8GB

The following table details the database virtual server used in the testing.

	Oracle Database
AWS RDS Instance Type	db.m5.2xlarge
CPU	8 Virtual Processors
RAM	32GB
DB Version	18c

4 User Experience Timings

4.1 Introduction

This section describes the work carried out to generate timing information for various SampleManager web client functions for use when comparing different network configurations.

4.2 Configuration

	Application Server	Web Server	Browser Client
AWS EC2 Instance Type	m5-4xlarge	m5-4xlarge	c5-xlarge
OS	Windows Server 2016 Datacenter 64-bit	Windows Server 2016 Datacenter 64-bit	Windows Server 2019 Datacenter 64-bit
CPU	16 Virtual Processors	16 Virtual Processors	4 Virtual Processors
RAM	64 GB	64GB	8GB

The following table details the application and database virtual server used in testing.

Oracle Database	
AWS RDS Instance Type	db.m5.2xlarge
CPU	8 Virtual Processors
RAM	32GB
DB Version	18c

4.3 Web Server Data

An application will perform differently dependent on configuration and data. It's important that the performance metrics are not done on a blank system. Prior to test execution each system was loaded with a known data set.

4.4 Network Performance

4.4.1 Introduction

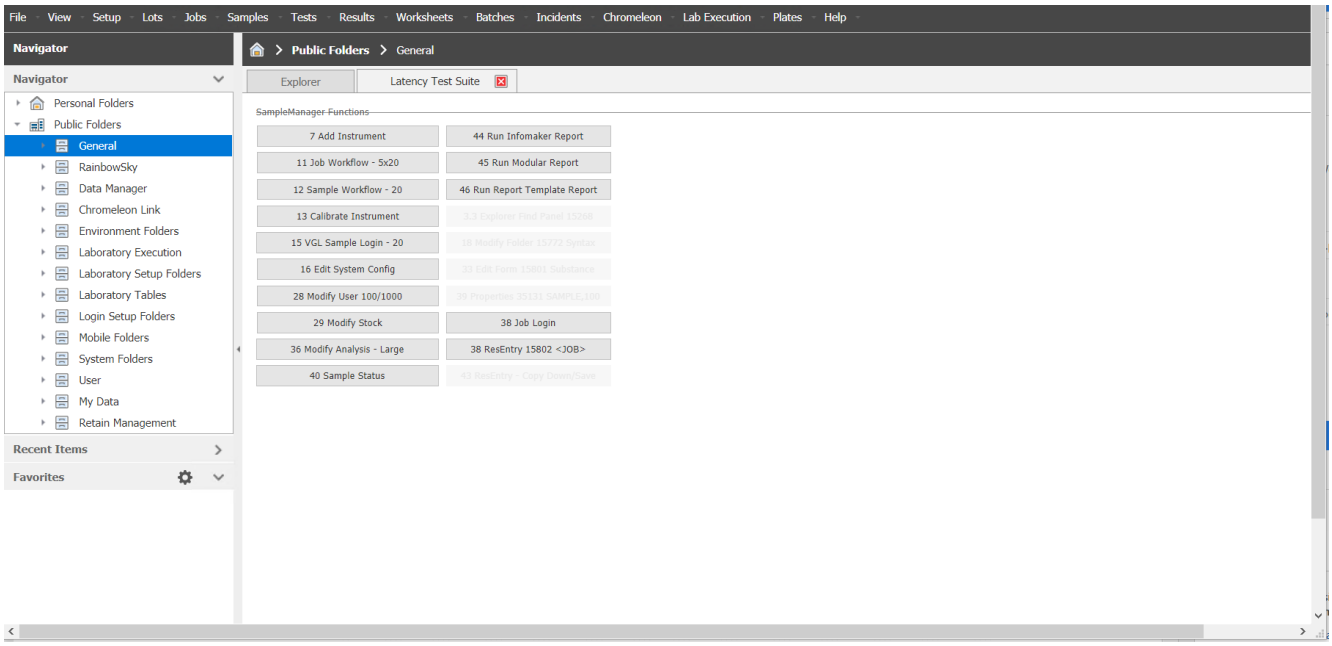
This section describes the work carried out to determine the effect of network latency on the usability of SampleManager web client.

4.4.2 Setup

It was necessary to develop a script to encompass a wide range of SampleManager functionality. It's not necessarily typical use scenarios, but rather a representative set of the different types of function within SampleManager web client. Essentially this is a cut down version of the User Experience script.

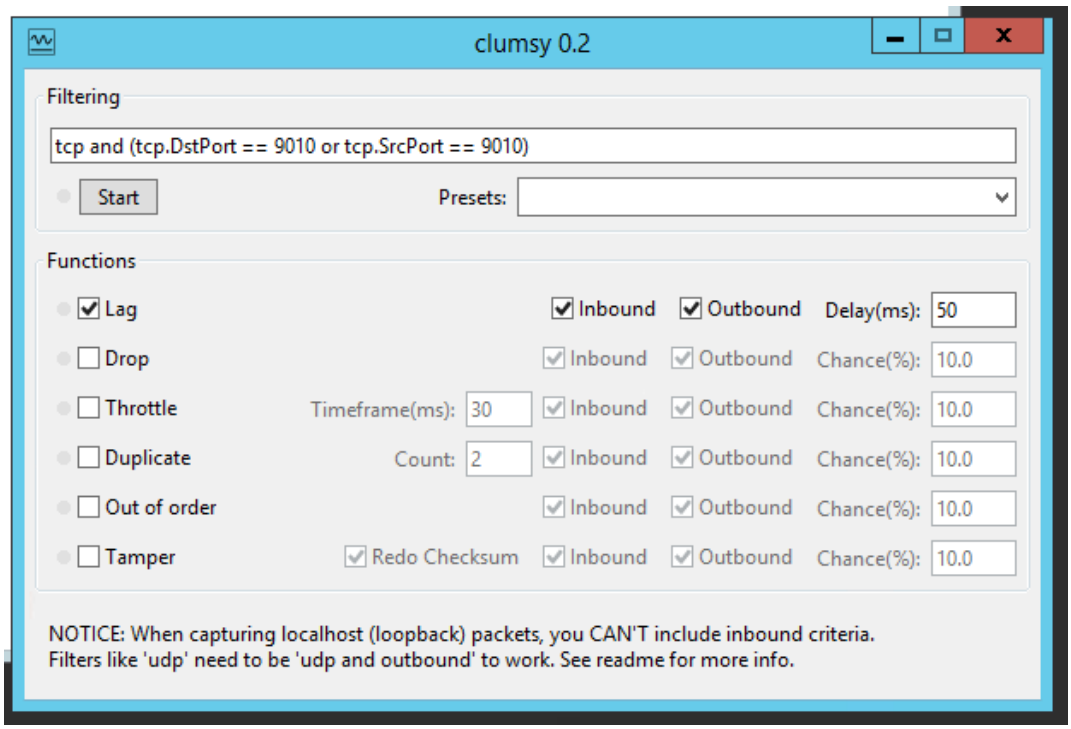
Step	Description	Type
1	Login to SampleManager/Explorer Loaded	System
2	Add Instrument	Static
3	Large Job Workflow Login. 5x20 Samples	Dynamic
4	Large Sample Workflow Login. 20 Samples	Dynamic
5	Calibrate Instrument	Static
6	Sample VGL Login. 20 Samples	Dynamic
7	Edit System Configuration	System
8	Modify Stock	Static
9	Modify Analysis (Large number components/lists)	Static

To simplify repeatability, the majority of these functions were added to a simple latency test suite form and executed via commands.



4.4.3 Monitoring

Clumsy was used to introduce latency on the network.



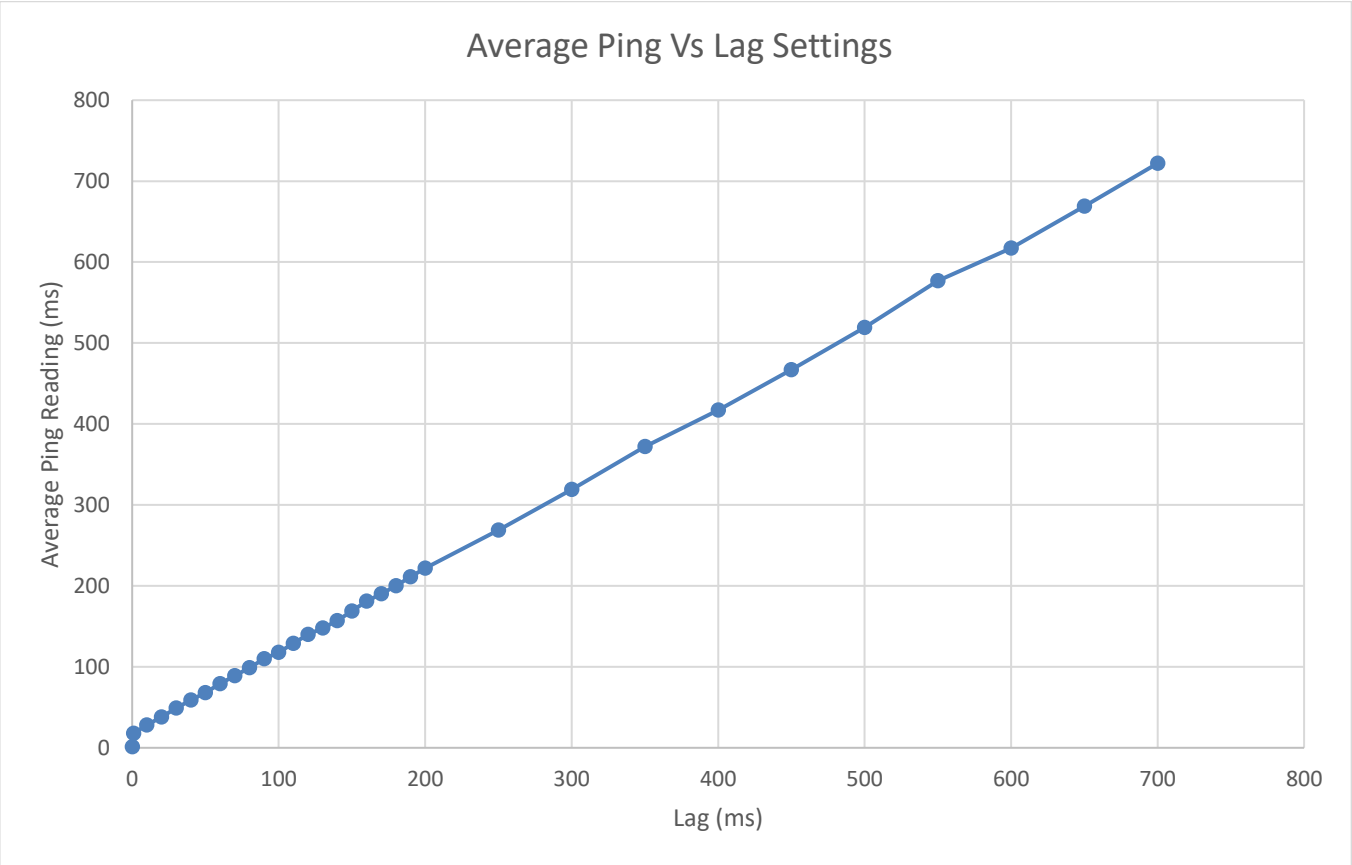
Two different scenarios were used:

1) Network latency between web server and browser client machine.
   tcp and (tcp.DstPort == 9010 or tcp.SrcPort == 9010)
2) Network latency between web server and application server.
   tcp and (tcp.DstPort == 56100 or tcp.SrcPort == 56100)

The lag setting on this tool was calibrated against the ping rate as follows:

Inbound Lag (ms)	Outbound Lag (ms)	Average Ping (ms)	Ping Reading (ms)			
0	0	1	1	1	1	1
1	1	18	11	33	18	12
10	10	28	37	21	37	19
20	20	38	40	35	36	41
30	30	49	34	58	40	65
40	40	59	43	70	51	75
50	50	68	84	68	49	73
60	60	79	64	88	72	94
70	70	89	96	78	101	83
80	80	99	116	98	79	104

90	90	110	124	103	118	97
100	100	118	125	105	133	112
110	110	129	135	117	144	123
120	120	140	136	157	142	128
130	130	148	130	157	139	166
140	140	157	145	170	146	170
150	150	169	177	156	181	163
160	160	181	198	182	161	183
170	170	190	188	209	192	171
180	180	200	192	208	186	214
190	190	211	198	210	228	209
200	200	222	209	231	212	237
250	250	269	282	257	279	258
300	300	319	314	338	323	304
350	350	372	358	382	363	387
400	400	417	419	438	415	399
450	450	467	461	486	471	450
500	500	519	508	531	504	533
550	550	577	587	571	584	568
600	600	617	602	626	608	635
650	650	669	666	690	671	652
700	700	722	739	720	704	725



The calibration result indicates that there is a delta of 20ms between lag setting in Clumsy tool and the actual network latency. Therefore, for example, in order to introduce 120ms latency between two machines for testing, a lag of 100ms should be set in Clumsy tool.

4.4.4 Application Tier Latency

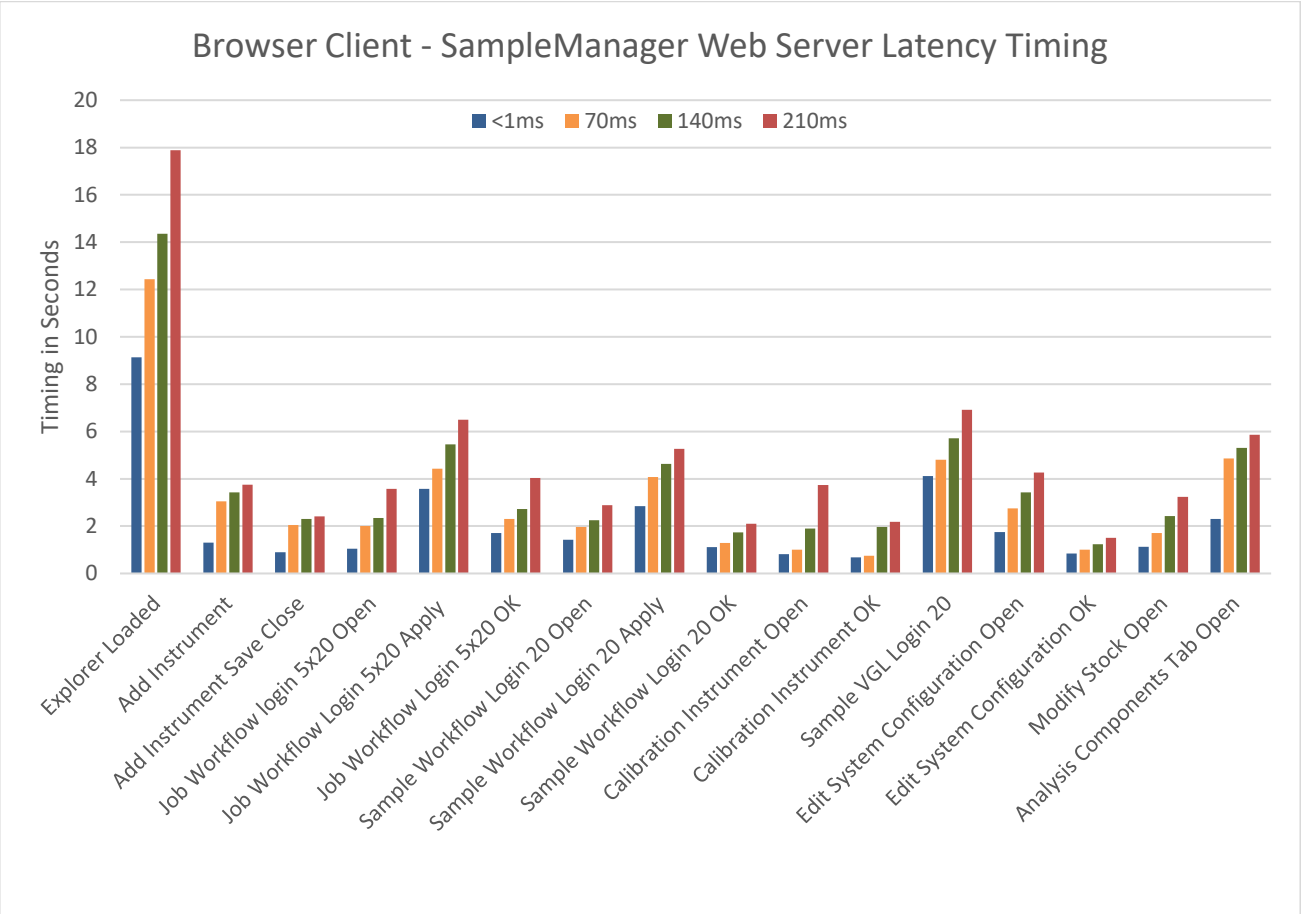
For screens with lots of UI elements, there is synchronization between the web server and application server, so as more elements are added, latency will affect the user experience. For screens where the majority of the work is creating data, for example saving result entry, these will be less affected.

For high latency environments, when customizing the system, consideration should be made to ensure that the number of round trips is minimized.

Timings were generated for <1ms, 70ms, 140ms and 210ms.

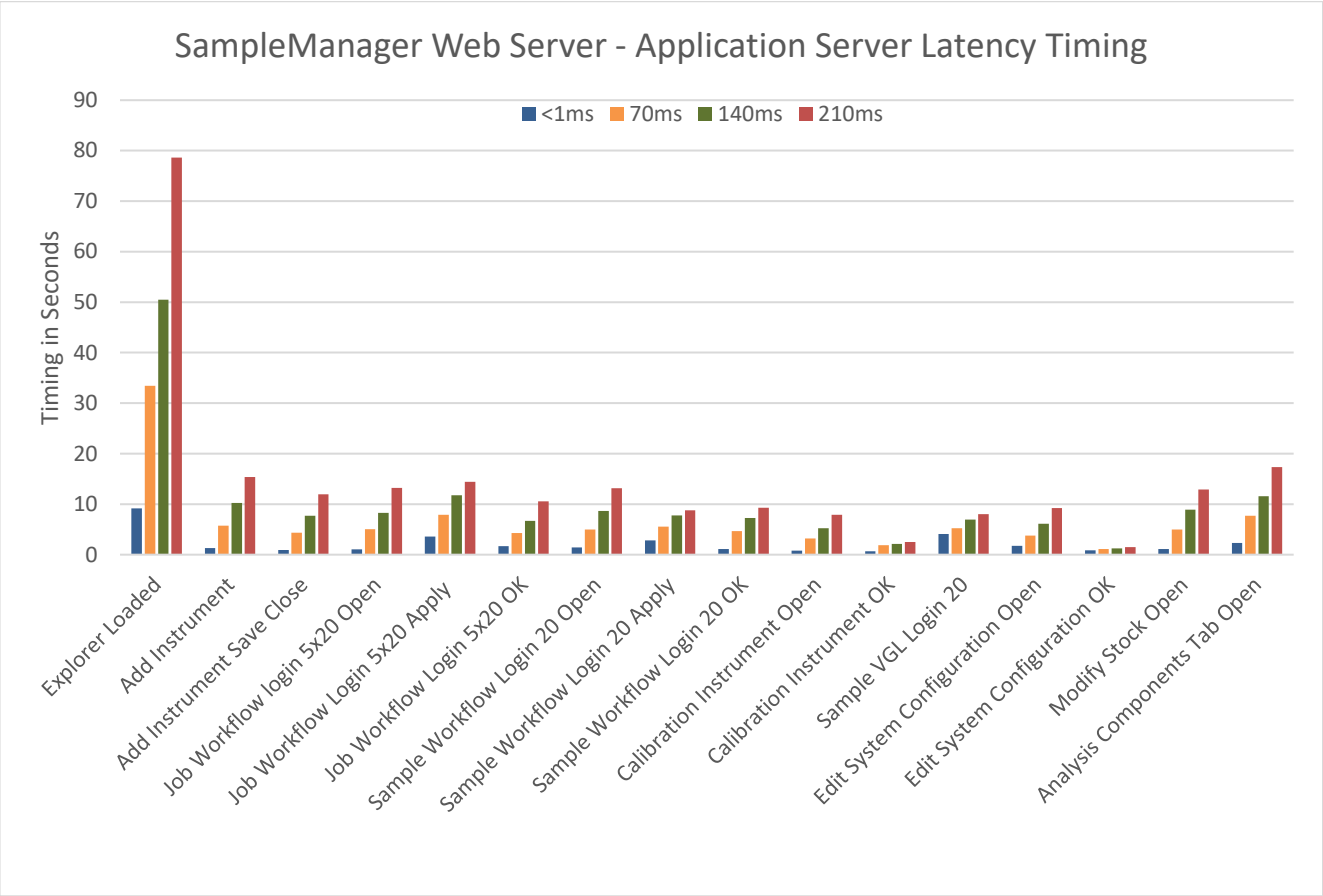
For mainly user interface operations there is quite a linear trend in timing versus latency.

1) Network latency between web server and browser client machine.



In summary – the higher the latency between the browser client and the web server, the longer it will take for screens to appear/update. The bigger impact is on system login and form loading times, whilst the impact on form closing is less significant. Opening Analysis Component Tab, Edit System Configuration and Modify Stock give useful indications of system performance at differing browser client to web server latencies. All in all, lower network latency between the browser and the web server is desired for optimal performance.

2) Network latency between web server and Samplemanager application server.



In summary – the impact of latency between the web server and the application server is very apparent on the time it will take for screens to appear/update, much more so than the latency between the browser client and web server. Latency in this tier also has a greater impact on the time to log in to the system. Explorer Loaded (login) is quite a useful indication of how the system will become sluggish as the latency increases.

4.5 Conclusions

1) SampleManager web server has higher demand on CPU than memory. The number of concurrent connections supported by a SampleManager web server with less than 8 virtual core processors is about 10 per 2 virtual core processors, while the number of concurrent connections supported by a server with greater than 16 virtual core processors is about 10 per 1 virtual core processor.

2) The largest impact network latency has is between the SampleManager web server and application server, instead of between the Samplemanager web server and the browser client machine.